QIYU DONG

Mathematics undergraduate | Optimization and computational mathematics
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EDUCATION

Hong Kong University of Science and Technology (HKUST) | Sep 2024–Present

BSc Mathematics (Pure Mathematics Track); declared Minor in Information Technology

Overall CGA: 3.436/4.3 (as of August 2026)

Selected grades: Discrete Structures (A+); Honors Linear and Abstract Algebra I–II (A/A−).

Programming with C++ (A−); General Biology I–II (A− each).

National University of Singapore (NUS) | Exchange student | Aug–Dec 2026

In progress: Non-Linear Programming; Statistical Learning I; Mathematics for Artificial Intelligence; Ordinary Differential Equations.

RESEARCH AND TECHNICAL PROJECTS

3D Container Loading Optimization

Independent project | Aug 2025–Present | C++, Python, OR-Tools CP-SAT
github.com/Dongqiyu829/3d-container-loading-optimization

- Developed a prototype for packing boxes into a container using heuristics and CP-SAT. The current Windows application lets users view and check loading plans.
- Planned work: compare packing quality and runtime across different box sets and improve the search strategy.

Quantum Groups and Computational Knot Invariants

Undergraduate Research / UROP | Supervisor: Prof. Ivan Ip, HKUST | Python
github.com/Dongqiyu829/Calculator-of-Knots-and-Links

- Studied quantum groups and Lie algebras and built a Python GUI for knot and link calculations. The project now includes R-matrix methods for computing invariants.

Initial prototypes used GitHub Copilot; later extensions used AI assistance.

TECHNICAL SKILLS

Programming: Python, C++; introductory PyTorch, Keras and TensorFlow.

Methods and tools: Optimization, heuristic search, constraint modeling, OR-Tools CP-SAT, Git/GitHub.

SELECTED ACHIEVEMENT

China Thinks Big Project | Global Finalist; National Second Prize | Data Analyst